

Education

- 2018–present Ph.D. in Physics, University of Massachusetts Amherst, USA
Advisors: Prof. Ben Heidenreich (HEP-th), (Expected: 2026)
- 2021 M.S. in Physics, University of Massachusetts Amherst, USA
- 2014–2018 M.Sc. in Physics, IISER Bhopal, India
- 2010–2013 B.Sc. in Physics (Honours), University of Delhi, India

Technical Skills

- Programming Julia (primary), Python, C++, Mathematica; experience with R
- Libraries ITensor, NumPy, SciPy, JAX, TensorFlow, Matplotlib, tensor-network libraries
- Methods Monte Carlo (worm algorithm, cluster algorithms, MCMC), tensor networks (DMRG/MPS), machine learning (normalizing flows, diffusion models)
- Tools Git, Linux/Unix, HPC clusters (SLURM), Jupyter, $\text{\LaTeX}$

Research Experience

- 2018–present **Ph.D. Researcher**, University of Massachusetts Amherst

Condensed matter and strongly correlated systems

- Investigate topological phases in low dimensional system on lattice using analytic and numerical tools like Tensor networks (primarily in Julia). One aspect involve study of boundary Renormalization group flows in Interacting majorana chain hosting symmetry-enriched Ising criticality using DMRG/MPS; identified a “universal scaling functions” characterizing flows between symmetry-enriched and trivial Ising criticality.
- Investigate superfluid transitions in 3D Bose gases with long-range Coulomb interactions using Monte Carlo simulations; measure superfluid stiffness, critical temperatures, and finite-size scaling to test universality.
- Analyzed dynamics of mobile impurities coupled to Dirac surface states of topological insulators using linked-cluster techniques; identified power-law decay and breakdown of the quasiparticle picture.

Algorithm development and high-performance computing

- Developed a new state of art Monte Carlo “Worm-algorithm” on classical lattice models for efficient handling of long interactions, and benchmarked against Cluster algorithm.
- Implemented complementary cluster algorithms for long range interaction to study superfluidity for related classical lattice models.
- Developed custom DMRG workflows utilizing tensor networks in Julia ITensor for computing energy, entanglement entropy, correlation functions, and other aspcts; optimized for HPC environments.
- Exploring physics-informed generative models (diffusion models, normalizing flows) and tensor cross-interpolation methods for accelerating sampling and high-dimensional function approximation.

String theory and quantum field theory

- Construct four-dimensional SCFTs from D3-branes at orientifolded toric Calabi–Yau singularities using geometric engineering and brane tilings, including the development of Lagrangian descriptions for strongly coupled 4d $\mathcal{N} = 2$ theories (such as the $R_{2,k}$ series, the rank-one E_6 Minahan–Nemeschansky theory, and the Argyres–Wittig theory).
- Analyze infrared phases of 4d $\mathcal{N} = 1$ $SU(N)$ gauge theories with tensor matter by introducing quad-CFT building blocks, using them to construct consistent dual descriptions and map out IR phase diagrams and symmetry enhancement patterns.
- Study “flavor orientifolds” of D3-branes at orientifolded toric Calabi–Yau singularities with non-compact O7-planes and flavor branes, showing how discrete torsion and Wilson-line data generate families of 4d $\mathcal{N} = 1$ SCFTs related by S-duality and distinct global-symmetry realizations.
- Use two-dimensional CFTs with conserved currents as controlled laboratories for the Weak Gravity Conjecture, investigating how discrete gauging and orbifolding reorganize charge lattices and twisted sectors and how modular invariance and defect operators restore consistency with lattice/sublattice WGC constraints.
- Investigate quad-CFTs from the perspective of mirror symmetry in type IIA string theory with D6- and O6-branes, relating symmetry enhancement and moduli-space structure of these non-Lagrangian theories to properties of the mirror Calabi–Yau geometry.

Publications

Refereed articles

- P1 **Deconfining $\mathcal{N} = 2$ SCFTs, or the Art of Brane Bending**, I. García Etxebarria, B. Heidenreich, M. Lotito, A. K. Sorout, *JHEP* 03 (2022) 140.
- P2 **Dynamics of Impurity in the Sea of Dirac Fermions**, A. K. Sorout, S. Sarkar, S. Gangadharaiah, *J. Phys.: Condens. Matter* 32, 415604 (2020).

Manuscripts in preparation and work in progress

- M1 **Boundary RG Flow in Symmetry-Enriched Ising CFTs**, A. K. Sorout, K. Wang, J. Zhang, manuscript in preparation (results presented at APS New England Meeting, Brown University).
- M2 **Superfluid Transition in a 3D Bose Gas with Long-Range Interactions**, A. K. Sorout, Chenan Wei, Nikolay Prokof'ev et al., manuscript in preparation.
- M3 **Worm and Cluster Monte Carlo Algorithms for Long-Range Interactions**, A. K. Sorout, Chenan Wei, and Nikolay Prokof'ev, manuscript in preparation.
- M4 **Deciphering IR Phases in 4d $\mathcal{N} = 1$ $SU(N)$ Gauge Theories Using Quad-CFTs**, B. Heidenreich, A. K. Sorout, manuscript in preparation (2025).(results presented at various conferences).
- M5 **Flavor Orientifolds and S-Duality in 4d $\mathcal{N} = 1$ SCFTs**, B. Heidenreich, M. Lotito, A. K. Sorout, manuscript in preparation (2025).(partial results presented at various conferences).
- M6 **Orbifolding/ Gauging Discrete Symmetries in 2d CFTs**, Ben Heidenreich, Nick Pittman, H. Rathnayak and A. K. Sorout, work in progress.
- M7 **Quad-CFTs and Mirror Symmetry**, Ben Heidenreich, Nick Pittman, and A. K. Sorout, work in progress.

Talks, conferences, and schools

- 2025 **APS New England Meeting**, Brown University, Providence, RI
Contributed talk: *Boundary RG Flow in Symmetry-Enriched Ising CFTs* – regional APS meeting highlighting current research in high-energy and condensed-matter physics.
- 2025 **String Phenomenology 2025**, Northeastern University, Boston, MA
Contributed talk: *Deciphering IR Phases of $\mathcal{N} = 1$ $SU(N)$ Gauge Theories Using Quad-CFTs* – international conference on string phenomenology and string compactifications.

2024 **Strings, Fields, and Deep Learning**

Contributed talk: *Flavor Orientifolds and S-Duality in $4d \mathcal{N} = 1$ SCFTs* – workshop at the interface of string theory, quantum field theory, and modern machine learning.

2023 **Prospects in Theoretical Physics: Understanding Confinement**, IAS / Princeton, NJ

Contributed talk: *Deciphering IR Phases of $\mathcal{N} = 1$ $SU(N)$ Gauge Theories* – advanced program on confinement and strongly coupled gauge dynamics.

2022 **APS April Meeting (High Energy Physics)**, New York City, NY

Contributed talk: *Deconfining $N = 2$ SCFTs, or the Art of Brane Bending* – national APS meeting covering broad topics in particle and gravitational physics.

Schools and Workshops

2025 **QNumerics: Numerical Methods in Quantum Information Science**, UMass Amherst

Summer school on high-performance quantum numerics, including GPU-accelerated simulation, tensor networks, Clifford-circuit simulation, and quantum-network modeling.

2024 **QNumerics: Numerical Methods in Quantum Information Science**, UMass Amherst (Mount Ida campus)

Summer school on numerical methods for quantum information, with emphasis on Julia-based simulation and practical algorithms for quantum systems.

2024 **IAIFI Summer School and Workshop**, Cambridge, MA

Program on AI and fundamental physics, with lectures and hands-on sessions at the intersection of machine learning and high-energy physics.

2024 **Physics Summer Workshop**, Simons Center for Geometry and Physics

Workshop on current topics in theoretical physics, including string theory, QFT, and related areas.

2023 **IAIFI Summer School and Workshop**

Summer program on machine-learning methods for fundamental interactions and data-driven approaches in high-energy physics.

2023 **Physics Summer Workshop**, Simons Center for Geometry and Physics

Workshop series on contemporary problems in high-energy and mathematical physics.

2023 **String Data Workshop**

Workshop on modern machine-learning approaches in string theory and related fields.

2022 **$5d \mathcal{N} = 1$ SCFTs and Gauge Theories on Brane Webs**, Simons Center for Geometry and Physics

Focused workshop on five-dimensional SCFTs, brane webs, and their gauge-theory realizations.

2019, 2024 **Theoretical Tests of the Swampland**, UMass Amherst

Workshops on swampland conjectures, quantum gravity constraints, and their implications for effective field theories.

Awards

2022–2023 **Distinguished Teaching Award**, University of Massachusetts Amherst

Campus-wide award for excellence in teaching; also nominated in 2021.

2014–2018 **MHRD Fellowship**, Government of India

National fellowship supporting M.Sc. studies at IISER Bhopal.